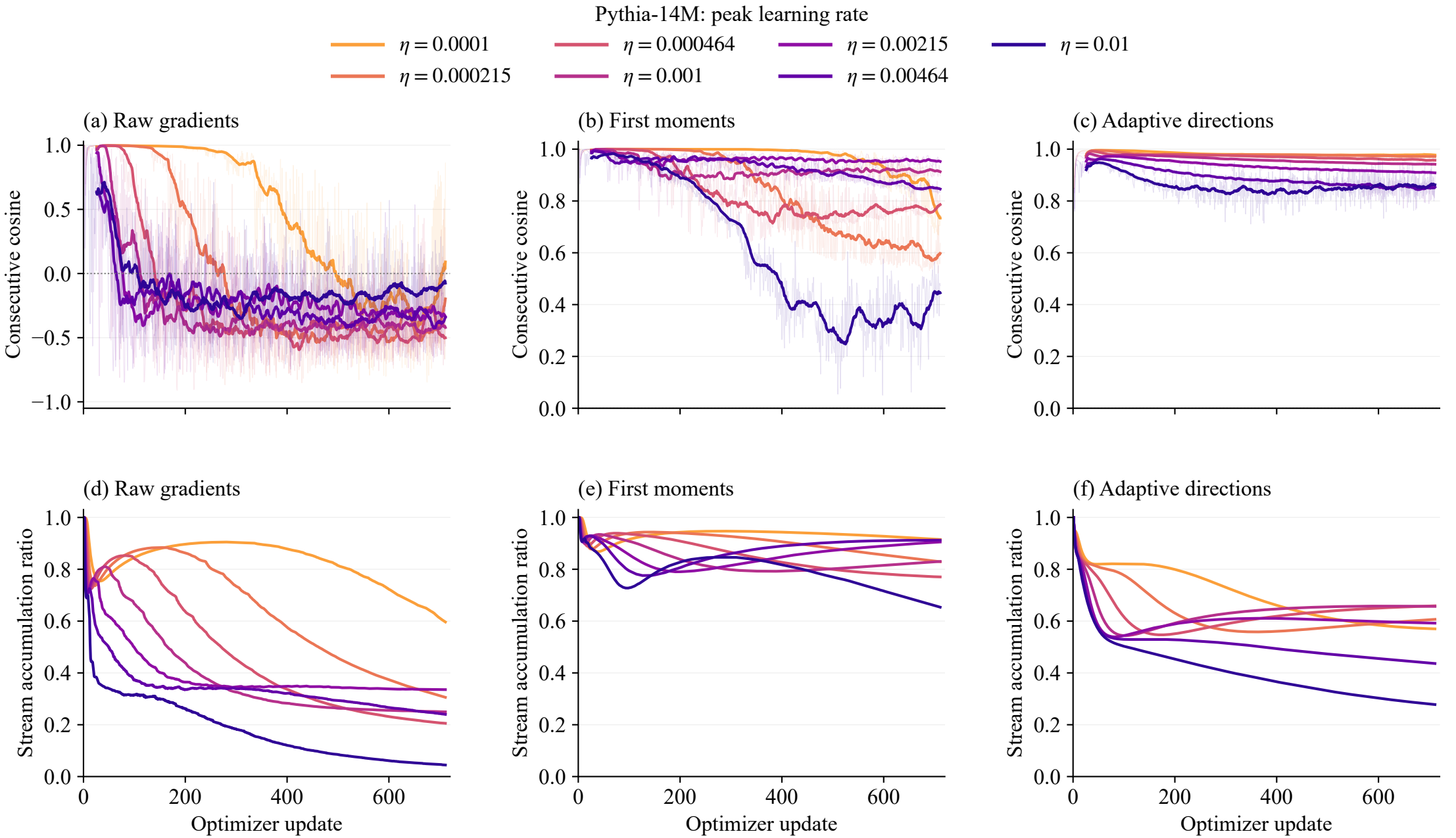

Pythia-14M: peak learning rate
$\eta = 0.0001$
$\eta = 0.000215$
$\eta = 0.000464$
$\eta = 0.001$
$\eta = 0.00215$
$\eta = 0.00464$
$\eta = 0.01$
(a) Raw gradients
(b) First moments
(c) Adaptive directions
(d) Raw gradients
(e) First moments
(f) Adaptive directions
Consecutive cosine
Stream accumulation ratio
Optimizer update
1.0
0.5
0.0
−0.5
−1.0
0.8
0.6
0.4
0.2
0
200
400
600